ADS401 HCI EVALUATION · GROUP H

Usability Evaluation of the XJTLU Museum AI System

A broader usability evaluation, including a 30-student post-visit experience experiment.

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kinger-ai.github.io

SAMPLE	GROUPS	OUTCOMES	ADJUSTED RESULT
30	**15 + 15**	**9 items**	**0 sig.**
XJTLU students, 17 undergraduate and 13 postgraduate.	AI-guide and no-AI comparison groups.	Q7–Q15 ratings from 1 to 5.	No Holm-adjusted p-value was below .05.

01 PROJECT CONTEXT

Overall topic: How usable is the museum AI guide, and how does AI-supported guidance shape the museum visit?

How do post-visit experience ratings differ between students who used the AI guide and those who did not?

This 30-person questionnaire is one embedded experiment, not the entire project.

02 DESIGN

1. Consent and participant information.
2. AI-guide or no-AI museum condition.
3. Visit the same exhibition context.
4. Complete the anonymous post-visit survey.
5. Compare Q7–Q15 ratings between groups.

Planned alternation by arrival did not appear as a strictly alternating recorded sequence. The study is not described as randomized.

03 MEASUREMENT

Engagement Absorption during the visit	**Interest** Interest maintained throughout
Learning Learned new knowledge	**Understanding** Deeper exhibit understanding
Reflection Prompted further thought	**Meaning** Visit felt meaningful
Context Connected exhibits to background	**Emotion** Emotional response to exhibits

04 ANALYSIS

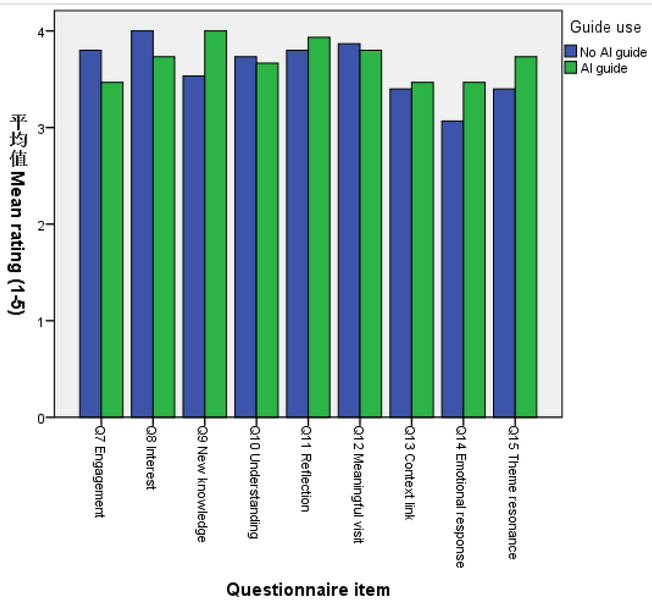
Two-sided independent-samples Mann–Whitney U tests for each ordinal item. Holm correction covers all nine comparisons. Rank-biserial correlation reports effect direction and magnitude.

15 / 15
No AI / AI participants

270 / 270
Valid Q7–Q15 ratings

No missing ratings, duplicate IDs, or out-of-range values were found.

05 RESULTS CHART



Blue: no AI guide. Green: AI guide. Bar height is descriptive; inference uses ranks.

06 KEY RESULT

5
Items with higher AI-group means

4
Items with higher no-AI means

Q9 had the largest descriptive difference: +0.47, raw p = .230, Holm p = 1.000, r_{rb} = +.249.

No item was significant before or after correction. This is absence of evidence for a difference, not proof of equivalence.

07 FULL RESULT PROFILE

ITEM	Δ AI−NO AI	RAW P	HOLM P
Q7 Engagement	−0.33	.278	1.000
Q8 Interest	−0.27	.328	1.000
Q9 New knowledge	+0.47	.230	1.000
Q10 Understanding	−0.07	.804	1.000
Q11 Reflection	+0.13	.757	1.000
Q12 Meaning	−0.07	.622	1.000
Q13 Context	+0.07	.875	1.000
Q14 Emotion	+0.40	.318	1.000
Q15 Resonance	+0.33	.427	1.000

08 INTERPRETATION

The current data do not provide sufficient evidence that AI-guide use changes any of the nine post-visit ratings.

- All outcomes are self-report.
- The 30-student sample is small.
- No verified primary endpoint was available.
- Allocation records prevent causal interpretation.

09 NEXT STUDY

1. Verify the allocation process.
2. Pre-specify one primary outcome.
3. Recruit a larger sample.
4. Keep the exhibition context comparable.

10 CONCLUSION

No significant difference in this sample. Do not claim equivalence or a causal AI effect. The main methodological lesson is that allocation records, a pre-specified endpoint, and correction for multiple tests must be planned before data collection.